

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures

COURSE CODE: CSA03

Register Number	192525464
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COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)*

Assessment Tool Weightage: 35%

Questions:2 EACH QUESTIONS CARRIES: 10 MARKS Total Marks:20 Duration: ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

*I _P.Harsha vardhan(192525464)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

P.Harsha Vardhan

Signature of the student:

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

ANSWERS

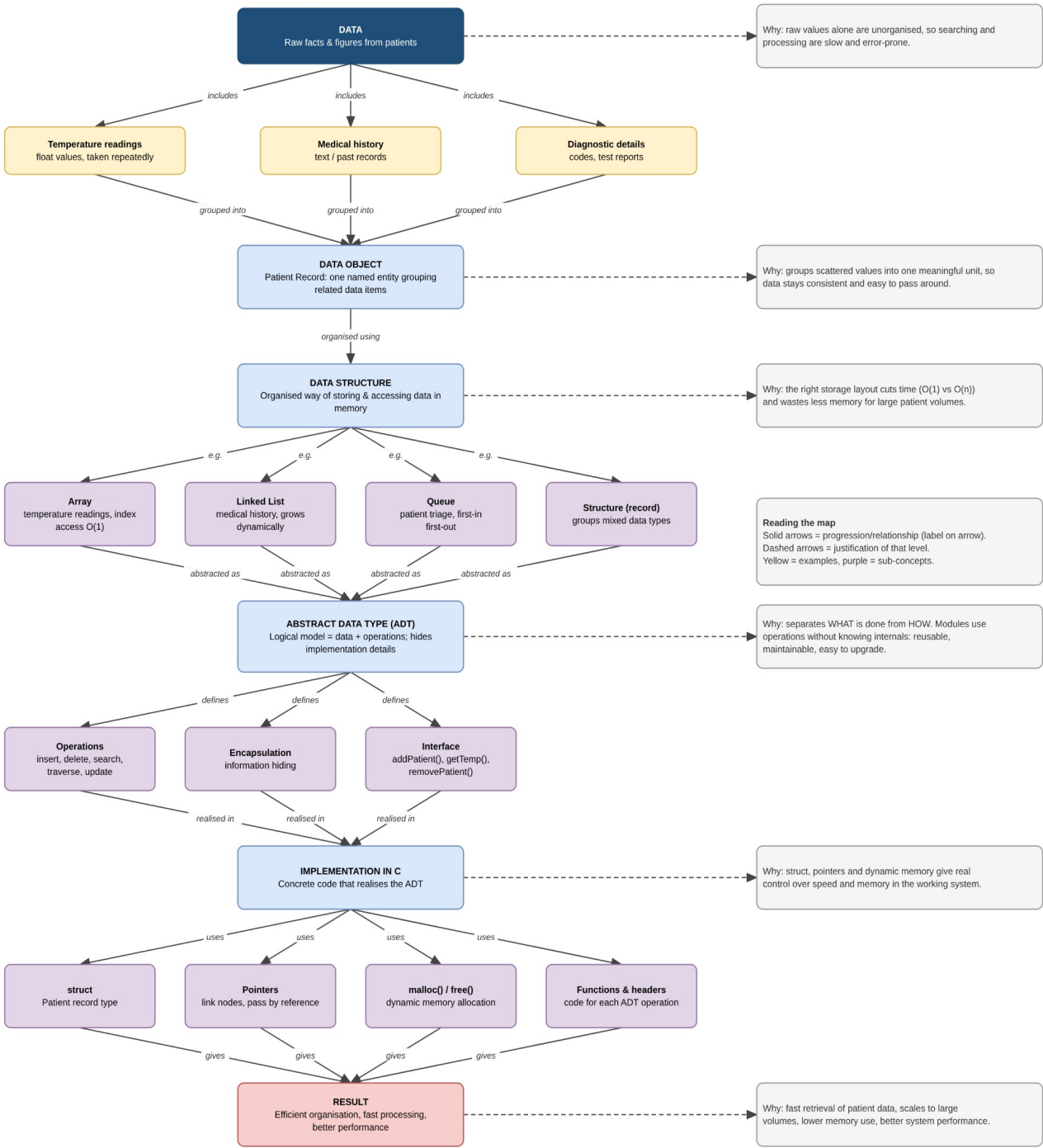
Q1.

Smart Healthcare System

Data → Data Object → Data Structure → ADT → C Implementation

- **Data:** Patient temperature, medical history, diagnostic details.
- **Data Object:** Patient record, sensor data, diagnosis record.
- **Data Structure:** Arrays, structures, linked lists, pointers.
- **ADT:** List, Queue, Stack.
- **C Implementation:** `struct`, pointers, functions, dynamic memory.
- **Benefit:** Faster access, efficient storage, reduced processing time, and better healthcare decisions.

CONCEPT MAP:



Q2.

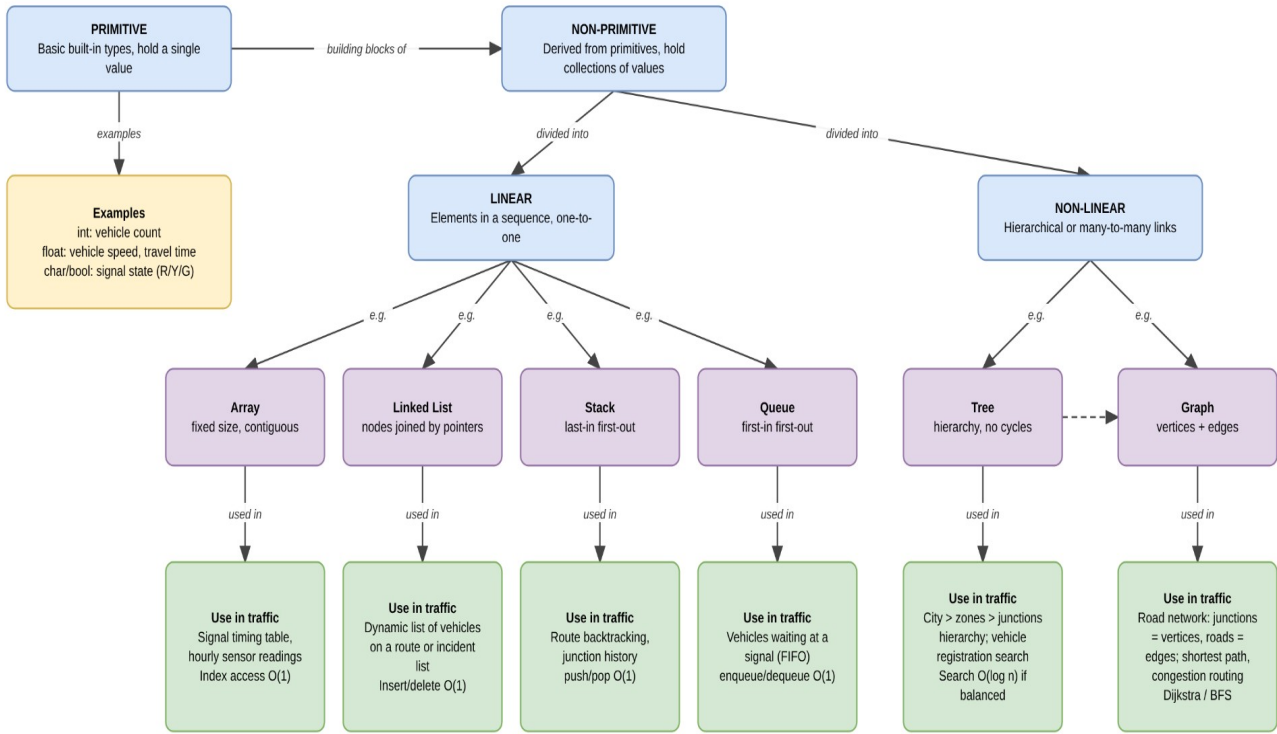
Smart City Traffic Management

Primitive → Non-Primitive → Linear / Non-Linear

- **Primitive:** Integer, float, character, Boolean.
- **Non-Primitive:** Arrays, linked lists, trees, graphs.
- **Linear:** Arrays, linked lists, queues, stacks — used for vehicle queues and traffic data.
- **Non-Linear:** Trees and graphs — used for road networks, route optimization, and shortest paths.
- **Benefit:** Efficient storage, faster processing, and improved traffic management.

CONCEPT MAP:

DATA STRUCTURES
Data in a smart city traffic system



Relationships between the structures

Primitive types are the building blocks: arrays hold ints/floats; list and tree nodes hold primitives plus pointers.
Array vs Linked List: contiguous memory vs pointer-linked nodes; either can implement Stack and Queue.
Tree is a connected graph with no cycles, so Tree is a special case of Graph (dashed arrow).
A Graph is stored using an array (adjacency matrix) or linked lists (adjacency list).

Justification of classification

Primitive vs non-primitive: single value vs a collection built from those values.
Linear: one predecessor/successor, simple sequential traversal $O(n)$, low memory overhead; suits ordered data like signal cycles and vehicle queues.
Non-linear: one element links to many; models real hierarchies and networks (zones, roads) and enables fast search (tree $O(\log n)$) and path finding (graph) at the cost of more memory and complexity.